

CERVICAL SPINE EXTENSION



Test Objective for the Cervical Spine Extension:

This movement tests the amount of available cervical spine extension.

How to Perform the Cervical Spine Extension:

The patient assumes the starting position by standing erect with feet together and toes pointing forward. The patient then looks up, aiming the face parallel with the ceiling. Use the line of their face as your reference line.

Functional Criteria:

- Line of the face within 10 degrees of the horizontal (80 degrees of cervical extension)
- Uniform Spine Curve
- Without excessive effort and/or loss of motor control

Common Compensations:

- Thorax/lumbar extension
- Head deviation right/left

Tips for Testing:

- Make sure the patient's mouth remains closed.
- Observe from the front and side.
- Do not coach the movement; simply repeat the instructions if needed.
- Was there pain?



CERVICAL SPINE FLEXION



Test Objective for the Cervical Spine Flexion:

This movement tests the amount of available cervical flexion.

How to Perform the Cervical Spine Flexion:

The patient assumes the starting position by standing erect with feet together and toes pointing forward. The patient then tries to touch the chin to the sternum, keeping the trunk erect during the movement.

Functional Criteria:

- Touch chin to sternum
- Uniform Spine Curve
- Without excessive effort and/or loss of motor control

Common Compensations:

- Thorax flexion
- Hinging from lower cervical spine

Tips for Testing:

- Make sure the patient's mouth remains closed.
- Observe from the front and side.
- Do not coach the movement; simply repeat the instructions if needed.
- Was there pain?



CERVICAL SPINE ROTATION



Test Objective for the Cervical Spine Rotation:

This movement tests the amount of available bilateral cervical rotation.

How to Perform the Cervical Spine Rotation:

The patient assumes the starting position by standing erect with feet together and toes pointing forward. The person rotates the head as far as possible to the right and then to the left. Use their nose-to-chin line as your reference line.

Functional Criteria:

- nose-to-chin line over mid-clavicle bilaterally (80 degrees of rotation)
- Without excessive effort and/or loss of motor control

Common Compensations:

- Cervical extension and/or sidebend
- Torso rotation
- Shoulder elevation

Tips for Testing:

- Make sure the patient's mouth remains closed.
- Observe from the front and side.
- Do not coach the movement; simply repeat the instructions if needed.
- Was there pain?



CERVICAL SPINE BREAKOUTS



Cervical Spine Pattern Breakout Rationale

Since the top-tier screen involves motion in all three planes for the cervical spine, there must be breakouts for each plane. This helps discover the primary dysfunction with each motion of the cervical spine. You only need to do the breakouts from the patient's dysfunctional top tier screen.

Cervical Spine Flexion SMCD

Since we are unable to remove body parts, our first step will be to reduce the postural stability requirements by having the patient lie supine with the head supported on a treatment table. In this position, the thoracic spine, shoulder girdle and cervical spine have minimal stability requirements. Instruct the patient to actively flex the neck. If standing cervical flexion is dysfunctional and/or painful, but the Active Supine Cervical Flexion Test is functional and non-painful, the patient has demonstrated full mobility. The diagnosis is a postural stability/motor control dysfunction affecting cervical flexion. This includes cervical spine, thoracic spine, and shoulder girdle postural dysfunction. Remember, this test is more difficult on the cervical spine than the standing top tier screen - if they show full range of motion, there is nothing wrong with the cervical spine for flexion.

If the Active Supine Cervical Flexion Test is dysfunctional and/or painful, check the same motion passively. If the Passive Supine Cervical Flexion Test is now FN, demonstrating full mobility, the patient has an active cervical spine flexion stability/motor control dysfunction. If passive motion is still limited or painful, we must check for a cervical flexion mobility dysfunction.

Occipitoatlantal & Cervical Flexion Mobility Dysfunction

In many cases, flexion dysfunctions occur between the occiput and the atlas - this can mimic pure cervical spine dysfunction. We continue the breakdown of the cervical spine flexion mobility dysfunction by assessing the occipitoatlantal (OA) joint. The Active Supine OA Flexion Test allows us to primarily isolate the OA joint (there is also contribution from the upper cervical spine) and check for 20 degrees of movement. If the Supine OA Flexion Test is functional and non-painful with bilateral rotation, the patient has a cervical spine(C2-C7) flexion mobility dysfunction. If the Passive Supine Cervical Flexion Test was an FP, this could also be an SMCD - you will need to perform segmental testing and soft tissue appraisal.

If OA flexion is dysfunctional, then there is an OA/Upper Cervical mobility dysfunction in conjunction with a possible cervical spine flexion mobility problem.

If there is pain with OA testing (FP or DP), note where the pain is and come back and examine the region in more detail after completing the entire SFMA.

Cervical Spine Rotation SMCD

As with cervical flexion, we are unable to remove body parts so we change the postural stability requirements by having the patient lie supine. Have the patient actively rotate their cervical spine bilaterally. If the Active Supine Cervical Rotation assessment is functional and non-painful, the patient has a postural stability/motor control dysfunction affecting cervical rotation.

CERVICAL SPINE BREAKOUTS



If the Active Supine Cervical Rotation Test is dysfunctional and/or painful, check the same motion passively. If the Passive Supine Cervical Rotation Test is now FN, the patient has an active cervical spine rotation stability/motor control dysfunction. If passive motion remains limited or painful, check for a cervical rotation mobility dysfunction.

Atlantoaxial and Lower Cervical Rotation Mobility Dysfunction

Half of cervical spine rotation comes from the atlantoaxial joint (C1-C2). We continue the breakdown of the cervical spine rotation mobility dysfunction by assessing C1-C2. The Supine C1-C2 Test allows us to isolate C1-C2 and check for 40 degrees of movement bilaterally. If atlantoaxial rotation is functional and non-painful, the patient has a lower cervical spine (C3-C7) rotation mobility dysfunction. If the Passive Supine Cervical Rotation Test was FP, this could also be a SMCD - you will need to perform segmental testing and soft tissue appraisal.

If atlantoaxial rotation is dysfunctional, then there is a C1-C2 mobility dysfunction in conjunction with a possible lower cervical rotation mobility problem.

If there is pain with C1-C2 testing, note where the pain is and come back and examine the region after completing the entire SFMA.

Cervical Spine Extension SMCD

Cervical extension is the last breakout for the Cervical Spine. This assessment allows two positional options to reduce the postural stabilization requirements. The first option is to have the patient lie supine with their head just off the end of a treatment table. The second option is to have the patient side lying with the healthcare practitioner supporting the patient's head. Either option allows the thoracic spine, shoulder girdle and cervical spine to have minimal stability requirements.

Passively, try to fully extend the neck. If the Supine Cervical Extension Test is functional and non-painful, but standing extension is poor, there is a postural stability/motor control dysfunction affecting cervical extension.

If the Supine Cervical Extension Test is dysfunctional, there is a cervical extension mobility dysfunction.

If there is pain with passive testing, note where the pain is and come back and examine the region after completing the entire SFMA.

CERVICAL SPINE BREAKOUTS



Selective Breakout Tests for Cervical Patterns:



**Active Supine Cervial
Flexion Test**



**Passive Supine Cervical
Flexion Test**



Supine OA Flexion Test



**Active Supine Cervical
Rotation Test**



**Passive Supine Cervical
Rotation Test**



**C1-C2 Cervical
Rotation Test**



**Supine Cervical
Extension Test**

CERVICAL SPINE BREAKOUTS



Limited Cervical Spine Patterns

Active Supine Cervical Flexion Test (Chin to Chest)



DN, DP or FP

FN

Passive Supine Cervical Flexion Test



FN

DN, DP or FP



There is a postural SMCD affecting Cervical Flexion. This includes cervical spine, thoracic spine and shoulder girdle postural dysfunction.

Active Cervical Spine Flexion SMCD

Active Supine OA Cervical Flexion Test (20°)

FN
Bilateral

DN

DP or FP

If Passive Supine Cervical Flexion (PSCF) was DP or DN then treat as Cervical Spine Flexion MD. If PSCF was FP can also be SMCD - perform segmental testing and soft tissue appraisal.

OA/Upper Cervical Flexion MD and/or possible Cervical Spine Flexion MD.

OA/Upper Cervical Flexion Pain/Dysfunction

Active Supine Cervical Rotation Test (80°)



FN

DN, DP or FP



There is a Postural SMCD affecting Cervical Rotation. This includes cervical spine, thoracic spine and shoulder girdle postural dysfunction.

Passive Supine Cervical Rotation Test (80°)

FN

DN, DP or FP



Active Cervical Spine Rotation SMCD

C1-C2 Cervical Rotation Test (40°)

FN

DP or FP

DN

Supine Cervical Extension

DN

DP or FP

FN

Cervical Extension MD

Cervical Extension Pain/Dysfunction

Postural and/or Active Cervical Extension SMCD

If Passive Supine Cervical Rotation (PSCR) was DP or DN then treat as Lower Cervical Rotational MD. If PSCR was FP can also be SMCD - perform segmental testing and soft tissue appraisal.

C1-C2 Cervical Rotation Pain/Dysfunction

C1-C2 MD and possible Lower Cervical Spine MD for rotation.

CERVICAL SPINE BREAKOUTS



Test Objective for the Active Supine Cervical Flexion Test:

To test the ability to actively flex the cervical spine in an unloaded posture.

How to Perform the Active Supine Cervical Flexion Test:

Instruct the patient to assume the supine position with the arms and hands by the thighs, legs relaxed. Have the patient lift their head and touch the chin to the sternum.

Functional Criteria:

Normal movement is bony part of the chin to the sternum.

Additional Information:

During the movement, make sure the patient's mouth remains closed. Do not allow scapular elevation or protraction throughout the movements. Also, watch for thoracic spine flexion as a compensation.

Possible Findings:

- FN: Actively brings chin to sternum, no pain
 - » If the finding is FN and the standing cervical flexion assessment was limited, there is a postural stability/motor control dysfunction affecting cervical spine flexion. This includes cervical spine, thoracic spine and/or shoulder girdle postural dysfunction.
- FP: Actively brings chin to sternum, pain
 - » Proceed to Passive Supine Cervical Flexion Test.
- DN: Inability to actively bring chin to sternum, no pain
 - » Proceed to Passive Supine Cervical Flexion Test.
- DP: Inability to actively bring chin to sternum, pain
 - » Proceed to Passive Supine Cervical Flexion Test.



CERVICAL SPINE BREAKOUTS



Test Objective for the Passive Supine Cervical Flexion Test:

To test the ability to passively flex the cervical spine in an unloaded posture to differentiate between a mobility and stability/motor control dysfunction.

How to Perform the Passive Supine Cervical Flexion Test:

Instruct the patient to assume the supine position with the arms and hands by the thighs, legs relaxed. Passively move the patient's head such that the neck flexes, moving the chin toward the sternum.

Functional Criteria:

Normal cervical flexion is bony part of the chin to the sternum.

Additional Information:

During the movement, make sure the individual's mouth remains closed. The chin should reach the sternum without the need for thoracic flexion. Shoulder should remain relaxed and on table.

Possible Findings:

- FN: Passively bring chin to sternum, no pain
 - » Indicates an active cervical spine flexion stability/motor control dysfunction
- FP: Passively bring chin to sternum, pain
 - » Proceed to the Active Supine Occipitoatlantal Cervical Flexion Test.
- DN: Inability to passively bring chin to sternum, no pain
 - » Proceed to the Active Supine Occipitoatlantal Cervical Flexion Test.
- DP: Inability to passively bring chin to sternum, pain
 - » Proceed to the Active Supine Occipitoatlantal Cervical Flexion Test.



CERVICAL SPINE BREAKOUTS



Test Objective for the Supine Occipitoatlantal Cervical Flexion Test:

To test Occipitoatlantal (OA) and upper cervical mobility in an unloaded posture.

How to Perform the Supine OA Cervical Flexion Test:

Instruct the patient to assume the supine position with the arms and hands by the thighs, legs relaxed. Passively rotate the patient's head as far to the right as possible - to their end range. Then, instruct the patient to do a chin tuck. Repeat the same sequence on the left side.

Functional Criteria:

Normal movement is 20 degrees of OA/upper cervical flexion with bilateral rotation.

Additional Information:

Make sure to passively rotate the head as far as possible and maintain this rotation while checking OA flexion. The test is not to bring chin to clavicle – just a head nod looking for 20 degrees of flexion.

Possible Findings:

- FN: 20 degrees of OA/upper cervical flexion bilaterally, no pain
 - » If the finding is FN bilaterally, the patient has a cervical spine flexion (C2-C7) mobility dysfunction - complete local biomechanical exam. If Passive Supine Cervical Flexion Test was FP this could also be a stability/motor control dysfunction - perform segmental testing and soft tissue appraisal.
- FP: 20 degrees of OA/upper cervical flexion bilaterally, pain
 - » Indicates pain with cervical spine flexion - complete local biomechanical exam.
- DN: Less than 20 degrees of OA/upper cervical flexion, no pain (unilateral or bilateral)
 - » Indicates an OA mobility dysfunction. There is also a potential cervical spine flexion mobility problem. Complete local biomechanical exam.
- DP: Less than 20 degrees of OA/upper cervical flexion, pain (unilateral or bilateral)
 - » Indicates a painful OA mobility dysfunction. Complete local biomechanical exam.



CERVICAL SPINE BREAKOUTS



Test Objective for the Active Supine Cervical Rotation Test:

To test the ability to actively rotate the cervical spine in an unloaded posture.

How to Perform the Active Supine Cervical Rotation Test:

Instruct the patient to assume the supine position with the arms and hands by the thighs, legs relaxed. Have the patient rotate the head as far to the right as possible. Instruct the patient to repeat the movement on the left side.

Functional Criteria:

Normal cervical rotation is 80 degrees bilaterally.

Additional Information:

If the patient's head falls into extension when in the supine position, it is appropriate to elevate their head with towels - or your hands - until the cervical spine is in a neutral axis. Be careful to not elevate into cervical flexion. Thorax should remain still through the test.

Possible Findings:

- FN: 80 degrees of active cervical spine rotation, no pain
 - » Indicates a postural stability/motor control dysfunction affecting cervical spine rotation. This includes cervical spine, thoracic spine, and shoulder girdle postural dysfunction.
- FP: 80 degrees of active cervical spine rotation, pain
 - » Proceed to the Passive Cervical Rotation Test.
- DN: Less than 80 degrees of active cervical spine rotation, no pain
 - » Proceed to the Passive Cervical Rotation Test.
- DP: Less than 80 degrees of active cervical spine rotation, pain
 - » Proceed to the Passive Cervical Rotation Test.



CERVICAL SPINE BREAKOUTS



Test Objective for the Passive Supine Cervical Rotation Test:

To test the ability to passively rotate the cervical spine in an unloaded posture to differentiate between a mobility and stability/motor control dysfunction.

How to Perform the Passive Supine Cervical Rotation Test:

Instruct the patient to assume the supine position with the arms and hands by the thighs, legs relaxed. Passively rotate the patient's head to the right, keeping the cervical spine in proper alignment. Repeat the movement on the left side.

Functional Criteria:

Normal cervical rotation is 80 degrees of rotation.

Additional Information:

Thorax should remain quiet through the test. Control for cervical extension.

Possible Findings:

- FN: 80 degrees of passive cervical spine rotation, no pain
 - » Indicates an active cervical spine rotation stability and/or motor control dysfunction.
- FP: 80 degrees of passive cervical spine rotation, pain
 - » Proceed to the C1-C2 Cervical Rotation Test.
- DN: Less than 80 degrees of passive cervical spine rotation, no pain
 - » Proceed to the C1-C2 Cervical Rotation Test.
- DP: Less than 80 degrees of passive cervical spine rotation, pain
 - » Proceed to the C1-C2 Cervical Rotation Test.



CERVICAL SPINE BREAKOUTS



Test Objective for C1-C2 Cervical Rotation Test:

To test C1 and C2 rotation mobility in an unloaded posture.

How to Perform the C1-C2 Cervical Rotation Test:

Instruct the patient to assume the supine position with the arms and hands by the thighs, legs relaxed. Passively flex the patient's neck, taking the chin to their sternum or to their end range cervical flexion. Then, while passively maintaining cervical flexion, instruct the patient to actively rotate the neck as far to the right as possible. Repeat with rotation to the left.

Functional Criteria:

Normal C1-C2 rotation is 40 degrees bilaterally.

Additional Information:

Make sure the individual is fully flexed to their end range cervical flexion and maintain this flexion through C1-C2 rotation.

Possible Findings:

- FN: 40 degrees or greater of C1-C2 rotation, no pain
 - » Indicates a lower cervical spine (C3-C7) rotation mobility dysfunction. If Passive Supine Cervical Flexion Test was FP this could also be a stability/motor control dysfunction - perform segmental testing and soft tissue appraisal.
- FP: 40 degrees or greater of C1-C2 rotation, pain
 - » Indicates pain at C1-C2 with rotation - complete local Biomechanical exam.
- DN: Less than 40 degrees of C1-C2 rotation, no pain
 - » Indicates a C1-C2 mobility dysfunction. There is also a potential lower cervical spine rotation mobility problem. Complete local Biomechanical exam.
- DP: Less than 40 degrees of C1-C2 rotation, pain
 - » Indicates a painful C1-C2 mobility dysfunction. Complete local Biomechanical exam.



CERVICAL SPINE BREAKOUTS



Test Objective for the Supine Cervical Extension Test:

To test extension of the cervical spine in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Supine Cervical Extension Test:

Instruct the patient to assume the supine position on a table, head extending beyond the end of the table, arms and legs relaxed. Passively extend the neck as far as possible. If this is uncomfortable for the patient, a side-lying position can be used by supporting the head and passively extending.

Functional Criteria:

Normal cervical extension is 80 degrees. Use the line of the face (forehead to chin) as the reference line.

Additional Information:

Look to see if the individual can get the line of their face within 10 degrees of the vertical.

Possible Findings:

- FN: Ability to extend the neck 80 degrees, no pain
 - » If the finding is FN, and standing extension is poor, there is a postural stability/motor control dysfunction affecting cervical spine extension.
- FP: Ability to extend the neck 80 degrees, pain
 - » Indicates pain with cervical spine extension - complete local Biomechanical exam.
- DN: Inability to extend the neck 80 degrees, no pain
 - » Indicates a cervical spine extension mobility dysfunction. Complete local Biomechanical exam.
- DP: Inability to extend the neck 80 degrees, pain
 - » Indicates painful cervical spine extension mobility dysfunction. Complete local Biomechanical exam.

Optional side-lying version

